

Endpoint	Age	Parameter	Rate	Source
	18-24		1.971	Bureau of the Census (1997, No.22)
	25-44		2.475	
	45-64		1.796	

NOTE: The incidence rate is the number of cases per person per year. Prevalence refers to the fraction of people that have a particular illness during a particular time period.

D.6.1 Acute Bronchitis

The annual rate of acute bronchitis for children ages 5 to 17 was obtained from the American Lung Association (2002b, Table 11). The authors reported an annual incidence rate per person of 0.043, derived from the 1996 National Health Interview Survey.

D.6.2 Chronic Bronchitis Incidence Rate

The annual incidence rate for chronic bronchitis⁴¹ is estimated from data reported by Abbey et al. (1993, Table 3). The rate is calculated by taking the number of new cases (234), dividing by the number of individuals in the sample (3,310), dividing by the ten years covered in the sample, and then multiplying by one minus the reversal rate (estimated to be 46.6% based on Abbey et al. (1995a, Table 1).

Age-specific incidence rates are not available. Abbey et al. (1995a, Table 1) did report the incidences by three age groups (25-54, 55-74, and 75+) for “cough type” and “sputum type” bronchitis. However, they did not report an overall incidence rate for bronchitis by age-group. Since, the cough and sputum types of bronchitis overlap to an unknown extent, we did not attempt to generate age-specific incidence rates for the over-all rate of bronchitis.

D.6.3 Chronic Bronchitis Prevalence Rate

We obtained the annual prevalence rate for chronic bronchitis from the American Lung Association (2010a, Table 4). Based on an analysis of 2008 National Health Interview Survey data, they estimated a rate of 0.0437 for persons 18 and older; they also reported the following prevalence rates for people in the age groups 18-44, 45-64, and 65+: 0.0315, 0.0549, and 0.0563, respectively.

D.6.4 Lower Respiratory Symptoms

Lower respiratory symptoms (LRS) are defined as two or more of the following: cough, chest pain, phlegm, wheeze. The proposed yearly incidence rate for 100 people, 43.8, is based on the percentiles in Schwartz et al. (Schwartz et al., 1994, Table 2). The authors did not report the mean incidence rate, but rather reported various percentiles from the incidence rate distribution. The percentiles and associated per person per day values are 10th = 0 percent, 25th = 0 percent, 50th = 0 percent, 75th = 0.29 percent, and 90th =

⁴¹ Please note that this endpoint is not regularly considered in U.S. EPA analyses (July 2018).

0.34 percent. The most conservative estimate consistent with the data are to assume the incidence per person per day is zero up to the 75th percentile, a constant 0.29 percent between the 75th and 90th percentiles, and a constant 0.34 percent between the 90th and 100th percentiles. Alternatively, assuming a linear slope between the 50th and 75th, 75th and 90th, and 90th to 100th percentiles, the estimated mean incidence rate per person per day is 0.12 percent. (For example, the 62.5th percentile would have an estimated incidence rate per person per day of 0.145 percent.) We used the latter approach in this analysis.

D.6.5 Minor Restricted Activity Days (MRAD)

Ostro and Rothschild (1989, p. 243) provide an estimate of the annual incidence rate of MRADs per person of 7.8.

D.6.6 Work Loss Days

The yearly work-loss-day incidence rate per 100 people is based on estimates from the 1996 National Health Interview Survey (Adams et al., 1999, Table 41). They reported a total annual work loss days of 352 million for individuals ages 18 to 65. The total population of individuals of this age group in 1996 (162 million) was obtained from (U.S. Bureau of the Census, 1997, No. 22). The average annual rate of work loss days per individual is 2.17. Using a similar approach, we calculated work-loss-day rates for ages 18-24, 25-44, and 45-64, respectively.

D.7 Asthma-Related Health Effects

Several studies have examined the impact of air pollution on asthma development or exacerbation. Many of the baseline incidence rates used in the health impact functions are based on study-specific estimates. The baseline rates for the various endpoints are described below and summarized in Table D-14. The prevalence of asthma is summarized in Table D-15.

Table D-14. Asthma-Related Health Effects Incidence Rates

Endpoint	Age	Parameter	Rate	Source
New Onset Asthma	0-4	Incidence	0.0234	Winer et al. (2012, Table 1 & Table 2)
New Onset Asthma	5-11	Incidence	0.0111	
New Onset Asthma	12-17	Incidence	0.0044	
New Onset Asthma	18-34	Incidence	0.0040	
New Onset Asthma, Women	35-44	Incidence	0.0051	
New Onset Asthma, Women	45-54	Incidence	0.0046	
New Onset Asthma, Women	55-64	Incidence	0.0059	
New Onset Asthma, Women	65+	Incidence	0.0039	
Asthma Exacerbation, Shortness of Breath, African American	8-13	Prevalence	7.40%	Ostro et al. (2001, p. 202)

Endpoint	Age	Parameter	Rate	Source
Asthma Exacerbation, Wheeze, African American	8-13	Prevalence	17.30%	
Asthma Exacerbation, Cough, African American	8-13	Prevalence	14.50%	
Asthma Symptoms, Shortness of Breath	5-12	Prevalence	18.50%	Lewis et al. (2013, p. 51)
Asthma Symptoms, Wheeze	5-12	Prevalence	19.40%	
Asthma Symptoms, Cough	5-12	Prevalence	30.10%	
Asthma Symptoms, Chest Tightness	5-12	Prevalence	12.70%	
Asthma Symptoms (Albuterol Use), Albuterol Use	6-17	Incidence	2.2	Rabinovitch et al. (2006, Table 1)
Upper Respiratory Symptoms (URS)	9-11	Incidence	124.79	Pope et al. (1991, Table 2)

D.7.1 New Onset Asthma

The annual rate of new asthma onset is estimated from Winer et al. (2012, Table 1 and Table 2). Winer et al., 2012 identify newly diagnosed asthma from the 2006-2008 Asthma Call-Back Survey (ACBS) and Behavioral Risk Factor Surveillance System (BRFSS) as individuals diagnosed by a doctor, or other health professional, within the 12 months prior to the surveys.

D.7.2 Shortness of Breath

To estimate the annual rate of new shortness of breath episodes among African-American asthmatics, ages 8-13, we used the rate reported by Ostro et al. (2001, p.202). To estimate the annual rate of new shortness of breath episodes among asthmatic children ages 5-12, we used the rate reported by Lewis et al. (2013, p.51).

D.7.3 Wheeze

The daily rate of new wheeze episodes among African-American asthmatics, ages 8-13, is reported by Ostro et al. (2001, p.202) as 0.076. We multiplied this value by 100 and by 365 to get the annual incidence rate per 100 people. To estimate the annual rate of new wheeze episodes among asthmatic children ages 5-12, we used the rate reported by Lewis et al. (2013, p.51).

D.7.4 Cough

The daily rate of new cough episodes among African-American asthmatics, ages 8-13, is reported by Ostro et al. (2001, p.202) as 0.067. We multiplied this value by 100 and by 365 to get the annual incidence rate per 100 people. To estimate the annual rate of new cough episodes among asthmatic children ages 5-12, we used the rate reported by Lewis et al. (2013, p.51).

D.7.5 Albuterol Use

The average number of albuterol inhaler actuations ('puffs') per day for an asthmatic child, age 6-17, is reported by Rabinovith et al. (2006, Table 1) as 2.2 'puffs' per child per day.

D.7.6 Upper Respiratory Symptoms

Upper Respiratory Symptoms are defined as one or more of the following: runny or stuffy nose; wet cough; burning, aching, or red eyes. Using the incidence rates for upper respiratory symptoms among asthmatics, published in Pope et al. (1991, Table 2), we calculated a sample size-weighted average incidence rate.

D.7.7 Asthma Population Estimates

In studies examining the association between air pollution and the development or exacerbation of asthma, often times an estimate of the percent of the population with asthma is required. Asthma percentages were obtained from an American Lung Association (2010b) report summarizing data from NHIS. Table D-15 presents asthma prevalence rates used to define asthmatic populations in the health impact functions.